

STRICT PAPER TRADING ONLY (\$0.00 CAPITAL AT RISK)

HEXAGONAL ARCHITECTURE (21 DOMAIN PACKAGES)

HISTORICAL TEST BASELINE: 4,260 TESTS (99.4% COVERAGE)

1. PRODUCT IDENTIFICATION & POSITIONING

Project ORION is a proprietary multi-tenant web application and REST API runtime designed for quantitative researchers, algorithmic prop desks, and institutional software evaluators.

- Core Value:** End-to-end quantitative pipeline taking strategies from mathematical formulation through walk-forward efficiency scoring to paper incubation.
- Mathematical Hygiene:** Validates algorithmic robustness via qualified **LeakageGuard** temporal checks and programmatic quality gate hurdles.
- Asset Scope:** Proposed transferable software and intellectual property asset scope; clean corporate separation with zero customer liability.

2. MODERN TECHNOLOGY STACK

- Backend Engine:** Python 3.11+, FastAPI (ASGI), Pydantic v2 schemas, SQLAlchemy 2.0 Async ORM (24 routers).
- Frontend SPA:** React 18, Vite, TypeScript, Tailwind CSS, TanStack Query, Lucide Icons (20 routes).
- Database:** PostgreSQL 16 with 15 linear Alembic migrations managing 29 relational tables.
- Cache & Broker:** Redis 7 with graceful in-memory degraded fallback for sessions and rate-limits.
- PaaS Infrastructure:** Render PaaS blueprint (~\$14/mo baseline configuration hosting estimate).

3. TEN IMPLEMENTED CORE CAPABILITIES

- 1. Deterministic Backtesting:** Sequential chronological bar traversal with trade-by-trade analytics and cent-level reproducibility.
- 2. Temporal Leakage Protection:** **LeakageGuard** enforces timestamp monotonicity; prevents indicator look-ahead bias.
- 3. Multi-Parameter Optimization:** Grid search exploring parameter surfaces with metric-based ranking (Sharpe/Sortino).
- 4. Walk-Forward Analysis (WFA):** Rolling In-Sample & Out-Of-Sample slices computing Walk-Forward Efficiency (WFE).
- 5. Automated Quality Gates:** Programmatic hurdles ($WFE \geq 0.50$, $DD \leq 15\%$, parameter stability) governing promotion.
- 6. Realistic Paper Execution:** **PaperExecutionAdapter** models spread, adverse slippage, and position netting (\$0 live risk).
- 7. Pre-Trade Risk Engine:** Dynamic evaluation of margin, leverage ceilings (1:100 default), and drawdown limits.
- 8. Multi-Tenancy & RBAC:** 7 organizational roles governing 41 permissions; isolated database queries via **TenantContext**.
- 9. Structured Audit Trail:** Immutable **AuditLogModel** logging acting user IDs, timestamps, and JSON metadata.
- 10. Declarative Cloud Blueprint:** Single-command 4-tier PaaS deployment via **render.yaml** with preDeploy migrations.

4. AUTHORITATIVE IMPLEMENTED SAAS SUBSCRIPTION MODEL (STRIPE TEST MODE)

Tier Name	Monthly Price	Paper Accounts	Daily Order Limit	Trading Workers	FX Pairs & Retention
Free Sandbox	\$0 / month	1 Paper Account	100 orders / day	0 Workers	4 Major Pairs 30-Day Retention
Pro Trader	\$99 / month	3 Paper Accounts	2,500 orders / day	1 Worker	12 Liquid Pairs 365-Day Retention
Business Prop	\$299 / month	10 Paper Accounts	50,000 orders / day	5 Workers	All Pairs 5-Year Data Retention
Enterprise	Custom Contract	Unlimited	Unlimited Capacity	Custom Allocation	Full Cross-Asset 7-Year Retention

* Commercial status: Pre-revenue technology asset. Paying customers, subscribers, and recurring revenue are **NOT ESTABLISHED IN REPOSITORY**. Stripe integration operates in Test Mode.

5. PROPOSED TRANSFERABLE SOFTWARE SCOPE

- Complete Source Code:** Full Git repository with clean linear commit history and zero credentials.
- Automated Test Harness:** Documented historical baseline of 4,260 automated tests with 99.4% line coverage.
- Database Architecture:** 15 linear Alembic migrations managing 29 tables; ~7.2s DR restore historical benchmark.
- 18 Closing Dossiers:** Master due diligence data room under **docs/acquisition/**.

6. BOUNDARIES & BUYER RESPONSIBILITIES

- Paper Invariant:** Strictly paper simulation; exactly \$0.00 live financial capital at risk.
- Worker Loop:** Autonomous worker disabled by default in cloud (**WORKER_ENABLED=false**).
- Buyer-Provisioned Accounts:** Buyer independently provisions Render PaaS, Stripe, TwelveData API key, and custom domain DNS.
- REST Architecture:** Stateless HTTP polling; no WebSocket push streaming endpoints.